# AXEL v2 — Phase 7 Product Requirements Document

## Purpose

Phase 7 builds the Crypto vertical on top of AXEL's deterministic core. It adds crypto-specific data, analysts, strategies, section risk controls, backtesting, paper trading and a promotion gate. Creative agents propose; deterministic controls decide whether anything can reach execution. The approved strategy scope includes funding-rate arbitrage, cross-exchange arbitrage and on-chain flow. The initial Crypto section uses no leverage and requires a rolling 24-hour UTC stop boundary.

## Scope

In scope: exchange adapters, market/funding data, on-chain data, crypto features, funding/on-chain/microstructure/regime analysts, crypto playbook, arbitrage detection, UTC risk boundary, section risk controls, backtesting, paper trading, monitoring and promotion. Out of scope: initial leverage, autonomous leverage changes, unsupported venues, unrestricted DeFi execution and global risk changes.

## Architecture

\[Exchange/Chain\ Data Normalization Features Analysts Strategy/Panel Section\ Risk Deterministic\ Core Execution\] All proposals remain subject to global risk, execution and Watchdog controls.

## Exchange Adapter

Each venue adapter exposes common interfaces for symbols/instruments, trades/OHLCV, order books where supported, funding rates, mark/index prices and (when enabled) account, position, order and fill state. Provider-specific schemas stay inside adapters. Records require venue, instrument, event time, source timestamp, ingestion time, schema version and provenance/hash.

## Market Data

Normalize spot and approved derivatives with instrument type, base/quote assets, venue, contract metadata and timestamps. Handle outages, gaps, duplicates, clock skew and stale data. Cross-venue prices must retain venue and timestamp.

## Funding Data

Store current/historical funding, interval, predicted funding where available, mark/index prices and venue. Features use only data known at decision time. Derived features include venue funding spreads, rolling funding statistics, z-scores, persistence and basis/mark-index relationships.

## On-Chain Data

Use an adapter boundary for approved providers. Store chain, block number/time, entity/address classification, metric/value, provider and retrieval metadata. Metrics and classifications are versioned. Unverified wallet labels are never treated as facts.

## Crypto Feature Contract

Feature rows contain instrument/entity, event time, availability time, venue/source, feature-set version, raw-source versions, value, missingness/quality and transformation version. Initial families: returns/momentum, realized volatility, liquidity/spread/depth, funding/basis, cross-exchange spreads, volume/flow, on-chain activity and market-regime context.

## Funding Arbitrage

Identify meaningful funding differentials only after accounting for funding timing, fees, spread, slippage, transfer/settlement costs, latency, liquidity and venue risk. A raw funding spread is not sufficient; net expected edge must remain positive after realistic costs.

## Cross-Exchange Arbitrage

Detect price discrepancies using synchronized timestamps and sufficient liquidity. Model both legs, fees, slippage, latency and settlement assumptions. A one-leg execution failure must enter an explicit repair/hedge/incident path; perfect simultaneous fills cannot be assumed.

## On-Chain Strategy

Signals may use approved accumulation/distribution, exchange inflow/outflow or other documented metrics. Every signal records metric definition, provider, block/time, classification version and confidence. On-chain evidence is contextual and does not automatically imply a trade.

## Crypto Analyst Layer

Initial analysts are: Funding Analyst; On-Chain Analyst; Market Microstructure Analyst; and Regime/Risk Analyst. They emit typed Signals/proposals through the existing trust boundary.

## Section Manager

The Crypto Section Manager checks approved strategy membership, venue/instrument eligibility and section constraints, and sends only typed proposals to the deterministic core. It may abstain.

## No-Leverage Rule

The initial Crypto implementation must operate without leverage. Configuration must reject leverage/margin parameters rather than ignore them. Any future leverage feature requires a separately reviewed risk and validation change.

## Rolling 24-Hour UTC Stop

Support a rolling 24-hour UTC loss/stop boundary with explicit reference equity, realized/unrealized P&L treatment, window/reset calculation, fees, missing-data behavior, halt behavior and recovery authorization. Breach deterministically halts new Crypto risk.

## Venue Risk

Track venue health, API latency, rejects, maintenance, stale data, settlement/withdrawal status where relevant and unexplained balance changes. A venue may be disabled independently.

## Backtesting

Backtests must include venue-specific data, funding payments, fees, spread/slippage, liquidity, latency, cross-exchange synchronization, outages/missing intervals, UTC windows and walk-forward OOS validation. Arbitrage tests model both legs rather than perfect simultaneous fills.

## Paper Trading

Before Crypto live activation, run paper execution from live market data without capital movement. Track hypothetical net edge after costs, leg completion, funding, stale-data events, venue failures and reconciliation.

## Promotion Gate

• Realistic-cost OOS backtests pass. • Trade count and robustness requirements are satisfied. • No leakage remains. • Paper evidence is complete. • Reconciliation is stable. • Venue reliability is acceptable. • No-leverage enforcement passes. • Rolling 24-hour stop tests pass. • Kill-switch/Watchdog tests pass. • Human review explicitly approves the strategy/venue combination. Promotion is strategy- and venue-specific; success of one strategy does not authorize all Crypto strategies.

## Testing

### Data

Duplicate/out-of-order events, clock skew, venue disagreement, missing funding, stale order books and on-chain late/reorg data where applicable.

### Strategy

Funding calculations, synchronization, fees, slippage, partial fills, one-leg failure and net-edge rejection.

### Risk

No-leverage, rolling 24-hour stop, venue halt, global kill-switch and recovery.

### Replay

Fixed snapshots reproduce identical signals and strategy decisions.

### Paper

Hypothetical orders/fills reconcile without capital movement.

## Observability

Expose venue health, data lag, funding updates, spread opportunities, candidate counts, rejected edge, liquidity, hypothetical P&L, fees/slippage, leg completion, on-chain freshness, section exposure, rolling loss, stop state and incidents.

## Security

Exchange credentials are execution-only and least privilege; withdrawal permissions should be disabled where avoidable. Secrets never enter prompts/logs. Agents cannot submit exchange orders directly. External API/on-chain content is untrusted data.

## Assignable Work Packages

## Developer Boundaries

Crypto developers must not weaken global risk, introduce leverage into the initial implementation, bypass Watchdog, expose exchange credentials to agents, or create direct Agent-to-exchange execution.

## Definition of Done

• Approved exchange adapters operate through common contracts. • Funding/on-chain data are point-in-time and provenance-linked. • Strategies calculate net edge after realistic costs. • One-leg execution failure is explicitly handled. • Crypto analysts produce typed signals. • No-leverage enforcement passes. • Rolling 24-hour UTC stop passes. • Backtests are OOS and cost-aware. • Paper activity reconciles. • Venue/incident monitoring is operational. • Promotion evidence is complete and human-approved.

## Acceptance Test

Load fixed multi-venue market/funding/on-chain fixtures. A funding spread is accepted only when synchronized, cost-adjusted edge is positive. A deliberately unprofitable spread is rejected. A one-leg failure enters the defined incident path. Any leverage request is rejected. The rolling 24-hour stop halts new Crypto risk when breached. Paper orders reconcile correctly. No agent process can submit an exchange order.

## Handoff

After promotion, Crypto may use the deterministic execution boundary. Any live activation remains subject to the global kill-switch, Watchdog and risk controls proven in the Stocks live phase.